



Stress detection based EEG under varying cognitive tasks using convolution neural network

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Abstract

One tool for promoting mental health is human stress detection through multitasks of electroencephalography (EEG) recordings. This study proposed a short-term stress detection approach using VGGish as a feature extraction and convolution neural network (CNN) as a classifier based on EEG signals from the SAM 40 dataset. This database was recently available and was collected from 40 patients using 32 channels to identify performance on four tasks including Stroop color-word test (SCWT), answering arithmetic problems, finding mirror-identical images, and relaxing. Each task took 25 s to complete and was then repeated three times to record three trials. This means that the total EEG data contain 480 signals for four tasks recorded using 120 trials per task. The primary objective of this research was to track the amount of short-term stress that patients experienced while they engaged in the four mental tasks. Moreover, the VGGish-CNN model is applied to the SAM 40 dataset using five stages including signal preprocessing, segmentation, filtration, spectrogram, and classification process. We compared the VGGish-CNN model and the VGGish model for stress-based EEG classification to determine the best classification accuracy. The proposed approach for stress detection is the preliminary study that achieved an accuracy of 99.25% using the VGGish-CNN model on the SAM 40 dataset. Next, k-fold cross validation is performed to verify the efficiency of the VGGish-CNN model. This study can advance the application of brain–computer interface (BCI) and its use to identify patterns in EEG data that invoke stress-related inferences to aid in the diagnosis of mental disorders. In the future, investigation of human stress using EEG data will be useful in neurorehabilitation.

Keywords Electroencephalography (EEG) · SAM 40 dataset · VGGish-CNN model · Brain–computer interface (BCI) · Convolution neural network (CNN)

1 Introduction

Psychological stress is a problem that has become common in contemporary society and poses serious obstacles to both the mental health of the individual and society as a whole [1]. High levels of stress increase the chance of suicide if stress is not recognized in time.

There are two phases of stress including acute and chronic stress. Acute stress is short-term stress which leads to increased heartbeats while chronic is long-term stress and most harmful which leads to stroke and hypertension [2]. Stress is caused by physical, psychological, and behavioral effects that may disturb the state of the brain, body functions, and immune system [3]. Generally, the stress of finding employment after graduation causes college students to worry about their future, and, in extreme circumstances, it may even be fatal [4]. Recently, the

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COVID-19 pandemic has increased stress levels among healthcare professionals and individuals [5, 6]. Also, mental health is affected by climate change, nutrition, and lifestyle. According to statistics from the American Psychological Association [7], stress levels are increasing in 2023 and it declared that more than one in four respondents (26%) suffer from increased stress. Therefore, stress detection is very important through standard methods such as questionnaires, facial expressions, and clinical tests such as magnetic resonance imaging (MRI) and EEG signals [8]. These standard methods have limitations since they might not catch stress events. However, EEG signals have become widely accepted for use in assessing stress levels and achieved high performance in stress detection based on brain signals [9]. Cognitive impairment in all cognitive areas is a core feature of neuropsychiatric disorders. BCI systems [10] are used in medical diagnosis for many neuropsychological diseases such as stress, epilepsy, Alzheimer's, stroke, Parkinson's, sleep disorders, and brain tumors as well as neurorehabilitation. EEG signals are used as input data and are interpreted or analyzed with computer commands to determine mental states [11]. Neuropsychological tests are clinical auxiliary activities that aid in the diagnosis of brain conditions. The measurement of mental stress using EEG signal analysis has been examined by psychologists through neuropsychological tasks to help in assessing stress levels in patients. Therefore, EEG signal analysis by BCI and artificial intelligence (AI) is a major factor in determining stress levels with commendable accuracy.

There are types of public databases for EEG-based stress detection such as DEAP (Database for Emotional Analysis using Physiological Signals) [12], SEED (SJTU emotion EEG dataset) [13], WAY-EEG-GAL [14], and EEGMAT [15]. However, the limitations of these accessible public datasets are their small size, biased coverage, noise, and low subject representation [16]. As a result, creating a broad range with a sizable public dataset is helpful for the advancement of stress recognition-based EEG signals. In the majority of trials, the scientists employed whole-channel EEG signals to collect the database. The accuracy of outcomes is affected by the whole-channel signal, which is not suitable for investigation. EEG channel-choosing technique is an active topic of research area because it is unknown which channels would be useful for the stress classification process. Therefore, it is necessary to build an acceptable EEG database for stress detection taking into account age, gender, environmental influence, and lifestyle.

Stress recognition based on EEG is a branch of AI including machine and deep learning methods like convolution neural network (CNN) [17]. Most studies of AI algorithms-based stress recognition using EEG databases

have confirmed that many factors are the reasons for low findings. These factors are signal preprocessing, feature extraction, classification algorithm, and stress type.

Doma and Pirouz [18] proposed stress detection on the DEAP database using machine learning algorithms which achieved an accuracy of between 55 and 75%. To increase the classification accuracy, Kumar et al. [19] suggested an emotion recognition system using machine learning on DEAP datasets with 94.95% accuracy. EEG signals were employed to distinguish between the stress state and the resting state by support vector machine (SVM) classifier with an accuracy of 94%, 85%, and 80% at varying degrees of arithmetic problems [20]. To increase the precision of the SVM classifier for detecting mental stress levels, five distinct algorithms are used to assess the various stress level types with a 96.36% accuracy rate [21].

Recently, a hybrid deep learning algorithm has performed well for stress recognition from EEG signals during mental arithmetic tasks on the EEGMAT database [24]. However, hybrid deep learning has complex computations due to extensive parameter adjustment and longer running times. The choice of hybrid combinations to demonstrate the model performance on the EEG dataset was a laborious effort. Agrawal et al. [25] introduced machine learning models to classify the three stress levels. Table 1 shows the related works of stress detection classifiers with their strengths and weaknesses.

In early research, the SAM 40 database [32] accumulated from 40 students consisting of the attention tests (i.e., SCWT, answering mathematical problems, finding symmetric mirror images, and relaxing) for short-term stress analysis on 32 channels based on EEG signals.

However, the challenge associated with the SAM 40 database remains under survey in the selection of classification methods. The objective of this research is to develop the accuracy of stress classification based on a deep learning model on the SAM 40 database for four stress tasks. This is extremely important given the potential of VGGish architecture to extract useful features for stress detection. This is the first study to achieve accurate stress detection on the SAM 40 dataset using a deep learning approach, specifically the proposed VGGish-CNN model that helps in mental health assessment, diagnosis of neuropsychological disorders, and neurorehabilitation.

The contribution of the suggested model is summarized as follows:

- (1) This is an empirical contribution to simply combine existing signal processing techniques and deep learning networks (VGGish and CNN) to apply them to a novel EEG database namely the SAM 40 database [32].

Table 1 Previous works related to EEG data for stress classification with their strengths and weaknesses

References	Classifiers	Accuracy	Strength	Weakness
[26]	Multilayer perceptron (MLP), long short-term memory (LSTM)	93.1% with the LSTM	It is used to assess the effect of viewing videos on mental stress using accessible EEG headgear with four electrodes	Private database limited number of electrodes
[27]	Five machine learning classifiers: KNN, NB, SVM, LR, and MLP	85.2% with SVM	Alpha asymmetry is employed as a diagnostic to classify long-term stress	Limited data size
[28]	KNN and SVM	90%	Select the EEG features that are related to dimension and spectrogram	Classification between stressed and non-stressed groups limited data size
[29]	Stockwell transform (ST) with KNN and LDA	91.1%	Classification of five levels of mental tasks	Five participants only
[30]	SVM with error-correcting output code (ECOC)	94.7%	Classification of three phases of stress	Male subjects only limited number of electrodes
[31]	Particle swarm optimization and SVM	92.3%	Choose the important features of the stress	High execution time

- (2) This proposed study outlines a deep learning approach based on the VGGish architecture [33] as a feature extraction integrated with the CNN classifier for stress detection through five steps including signal preprocessing, segmentation, filtration, spectrogram, and classification process to obtain high test accuracy in stress classification.
- (3) To validate the results, the test setups represent the addition of new experiments for leave-out-subject classification using multiple runs (k-fold) to achieve a more practical and challenging evaluation of the generalizability of this model.
- (4) In this proposed concern, the first use of Mel spectrogram to analyze EEG signals was created and the first use of VGGish for feature extraction was applied to EEG signals for stress detection.
- (5) A comparative analysis was also performed to predict four stress tasks associated with or without the CNN model using five performance metrics.

2 Material and methods

2.1 Mathematical model-based stress detection from EEG signal

In this work, the suggested model is used for stress detection based on EEG signals from the SAM 40 database [32] as shown in Fig. 1. The mathematical model's steps are as follows:

- 1- Using a raw EEG signal from the SAM 40 database that consists of a matrix of 3200 data points $\times$ 32 channels [32].
- 2- Initially, the raw database of EEG signal-32 channels can go into the signal preprocessing step including normalization and construction in one channel to convert into audio signals with matrix 3200×1 based on time–amplitude representation.
- 3- The signal segmentation step is used to split the preprocessed signal into small segments in the time domain using a 25-ms periodic Hanning window function and then fed into the discrete Fourier transform (DFT) [34] to translate a domain's setting from time to frequency.
- 4- After that, the Mel-scaled filter bank [35] is calculated on audio signals to convert frequencies into Mel-scale which is used to significantly lower the resolution of the information and increase the strength of audio signatures.
- 5- Mel spectrogram [36] is applied to convert the audio signal into a 2D image which represents the time–frequency domain. Then, Mel spectrogram images are segmented with the size of 96×64 pixels, where 96 represents the number of Mel spectrograms, and 64 represents the number of Mel bands.
- 6- Mel spectrogram images are employed as input for the classification process. Finally, the classification process consists of 4096 features extracted from the VGGish [33] followed by a CNN classifier. Mel spectroscopy images were separated into 70% for training, 15% for validation, and 15% for testing. Next, the output of the CNN classifier is tested by the model's effectiveness metrics on the test dataset.

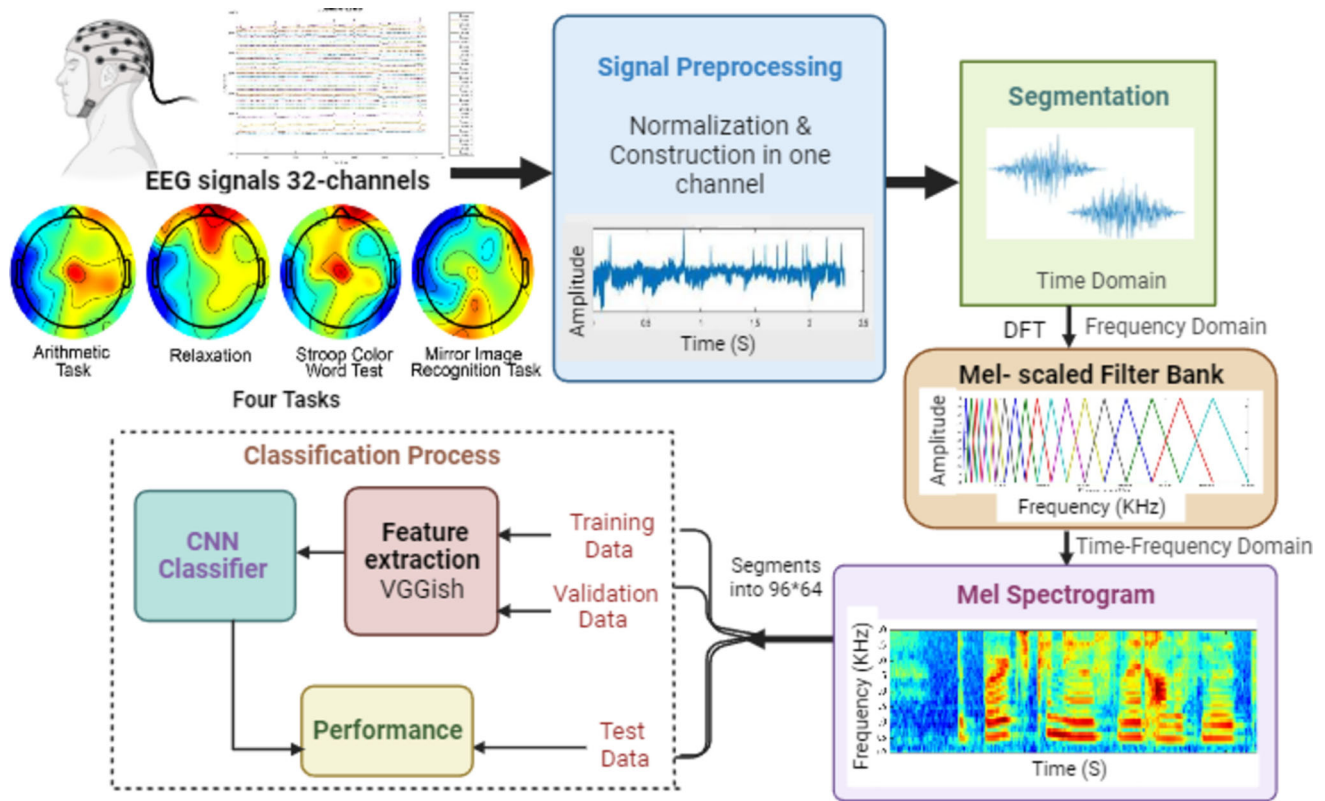


Fig. 1 Overview of EEG-based stress detection system using VGGish feature extractor and CNN classifier

2.2 SAM 40 Database

The recorded EEG signals from the SAM 40 database were adopted by Ghosh et al. [32]. This database was gathered from the 32-electrode Emotiv EPOC Flex Gel kit [37] as a BCI headset to monitor the EEG signal for stress detection during multitasking. This SAM 40 database is represented with a frequency of 128 Hz per second in both raw and filtered formats.

This study used the raw SAM 40 database from 32 channels that were obtained from respondents while they were engaged in four cognitive activities, namely SCWT, answering mathematical problems, finding symmetric mirror images, and relaxing as shown in Table 2. The EEG data contain 120 signals per task with a matrix of 3200×32 , where 3200 represents data points and 32 represents channel numbers as shown in Fig. 2.

SCWT [38] is a cognitive test that is used to assess a patient's ability to conclude a set of stimuli. The goal is to get the patients to recognize the colors painted on various colored panels. There are two situations including compatible color and incompatible color. The mirror image task [39] is a stress test that asks to decide if mirror images are symmetrical or asymmetrical. A mathematical task [40] is a mental test that asks if the mathematical problem solutions are correct or wrong.

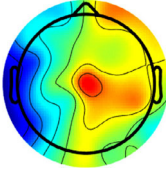
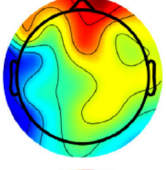
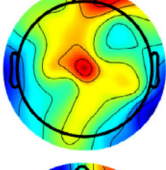
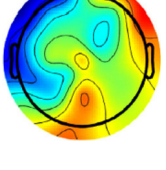
In this SAM 40 data, the guidelines for each task are shown for 10 s, where each task is carried out for 25 s with three trials.

2.3 Signal preprocessing

The EEG preprocessing strategy [41] significantly reduces the training time during the classification process, and interestingly, it greatly affects the classifier performance.

Initially, signal preprocessing of raw EEG-32 channels from the SAM 40 database [32] is performed using two steps namely normalization [42] and EEG construction in one channel. Along with the raw EEG data from the SAM 40 database, it consists of a matrix of 2300×32 , where 2300 represents data points and 32 represents channels. The function of normalization is to remove signal noise because this SAM 40 data are picked up from a BCI headset that is not an accurate representation of the signals originating from the brain. The function of EEG construction in one channel is to convert the recorded raw data with 32 channels to one-channel signals with a matrix of 2300×1 , where 2300 represents data points and 1 represents one channel. The graphical representation of EEG construction from the SAM 40 database with 32 channels to one channel for four stress tasks is depicted in Fig. 3. In this study, all 480 EEG signals including four stress tasks

Table 2 Description of EEG recordings from SAM 40 database

Stress Tests	EEG topography diagram	Function	No. of signals
Arithmetic		It is used to determine the arithmetic stimuli by the answer response on the screen. It is based on the correct or wrong solution of arithmetic problems	120
Relaxing		Firstly, a relaxation state is given to the students for 25 s while soothing music is played. Then, relaxation for 5 s is used after each task	120
SCWT		It is used to determine the difference in reaction times between congruent and inconsistent stimuli. It is based on a mismatch between a color's name and the printed color	120
Mirror image recognition		It is used to determine whether the reflected image is an image of itself. It is based on whether the reflected image is identical or different from the original one	120
Total			480

from the SAM 40 database are preprocessed to enhance the quality of the recorded signal. In Fig. 3, EEG construction in one channel is represented by the x-axis for time and the y-axis for amplitude.

2.4 Segmentation

EEG segmentation [43] of the entire set of observed time series is an important step to get the signal segments with relatively constant characteristics. There are numerous uses for EEG segmentation in the medical disciplines, including the identification of neurological diseases [44].

In this study, the preprocessed signal was divided into small segmented signals in the time domain and convert it to audio signals in the frequency domain at a frequency of 16 kHz by using DFT [34] as displayed in Eq. 1.

$$X(K) = \sum_{n=0}^{N-1} x(n) \exp\left(\frac{-j\pi m(n-1)}{N}\right) \quad (1)$$

$0 \leq n$, and $k \leq N - 1$

where $x(n)$ denotes the signal segment, $X(K)$ denotes the transformed sample (signal spectrum), n denotes the time

domain, m denotes the frequency domain, and N denotes the length of a signal segment.

The preprocessed EEG data are split up into signal segments of 25 m seconds, with each segment assuming 400 sample points for 99% overlap of the window's length and a hop size of 10 ms that is based on the Hanning window function [45] as shown in Eq. 2

$$\omega(n) = \frac{1}{2} \left[1 - \cos\left(\frac{2\pi(n-1)}{N}\right) \right], 0 \leq n \leq N-1 \quad (2)$$

where N is the Hanning window function's length and n denotes the window function's overall length.

2.5 Mel-scaled filter bank

The Mel-scaled filter bank [46] is a filter library that is employed successfully in the speech analysis model which is built upon pitch recognition.

It consists of several filters in the triangular form that apply the Mel frequency scale to convert an audio signal to discrete frequency bands which simulate the nonlinear human perception of sound. The Mel-scale attempted to emulate the nonlinear perception of audio by the human ear which is more sensitive at lower frequencies and less

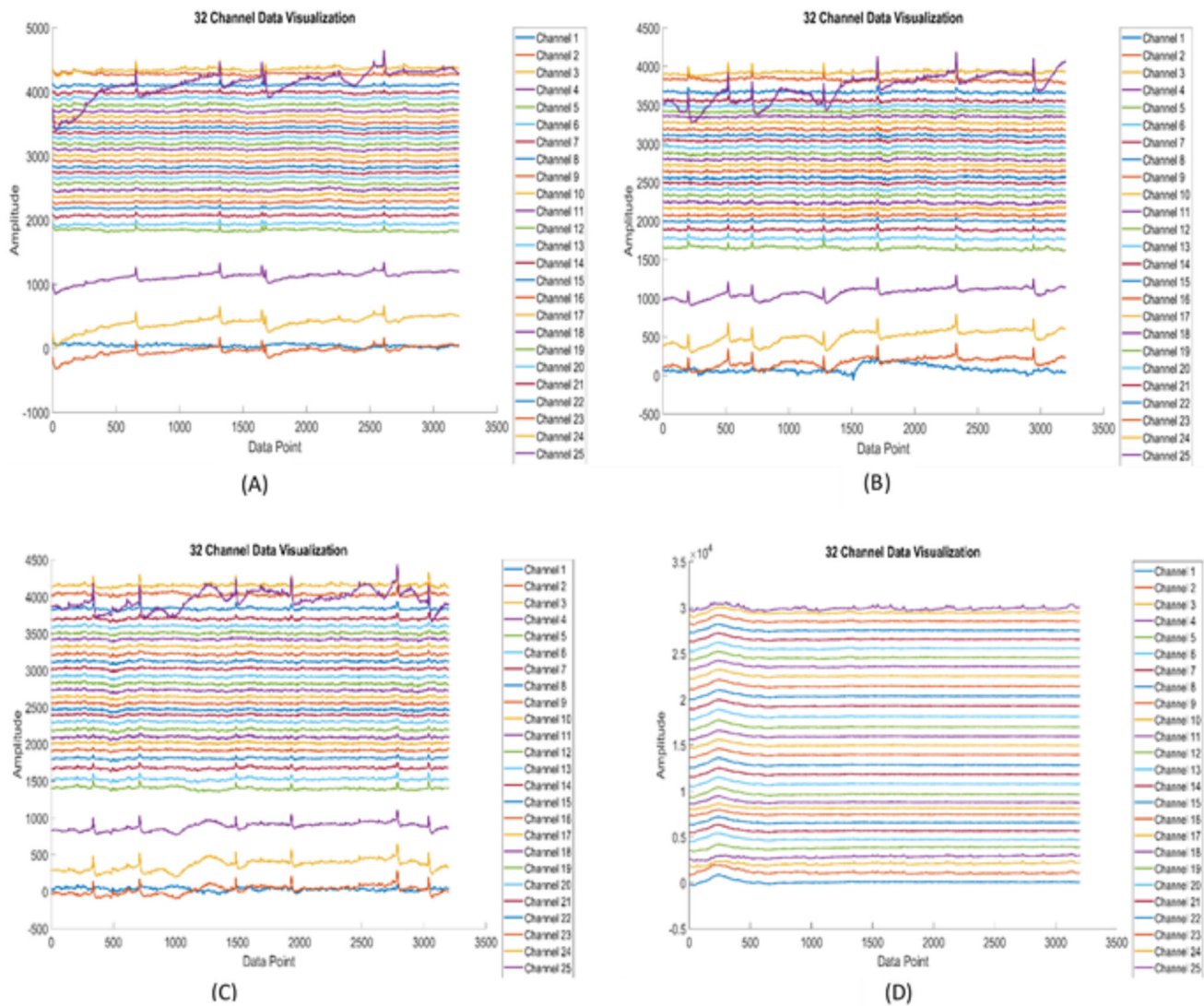


Fig. 2 EEG signals of four stress tasks from the SAM 40 database: **A** arithmetic, **B** SCWT, **C** mirror image recognition, and **D** relaxing

sensitive at higher frequencies. Eq. 3 is used to convert from f (frequency) to m (Mel-scale).

$$m = 2595 \log_{10}(1 + f/700) \quad (3)$$

The Mel-scaled filter bank is represented by the x-axis for frequency, while the y-axis for amplitude. The output of the Mel-scaled filter bank is computed by the product of signal spectrum $X(k)$ and filter $H_i(k)$ as shown in Eq. 4.

$$X_i = \sum_{k=0}^{N-1} H_i(k) |X(k)|^2 \quad (4)$$

where k represents the DFT from zero frequency to the Nyquist frequency [47].

In this study, each signal segment of 25 m seconds was passed into the Mel-scaled filter bank to choose the most accelerated parameters or to characterize the relationship between filter bank record spectral sizes.

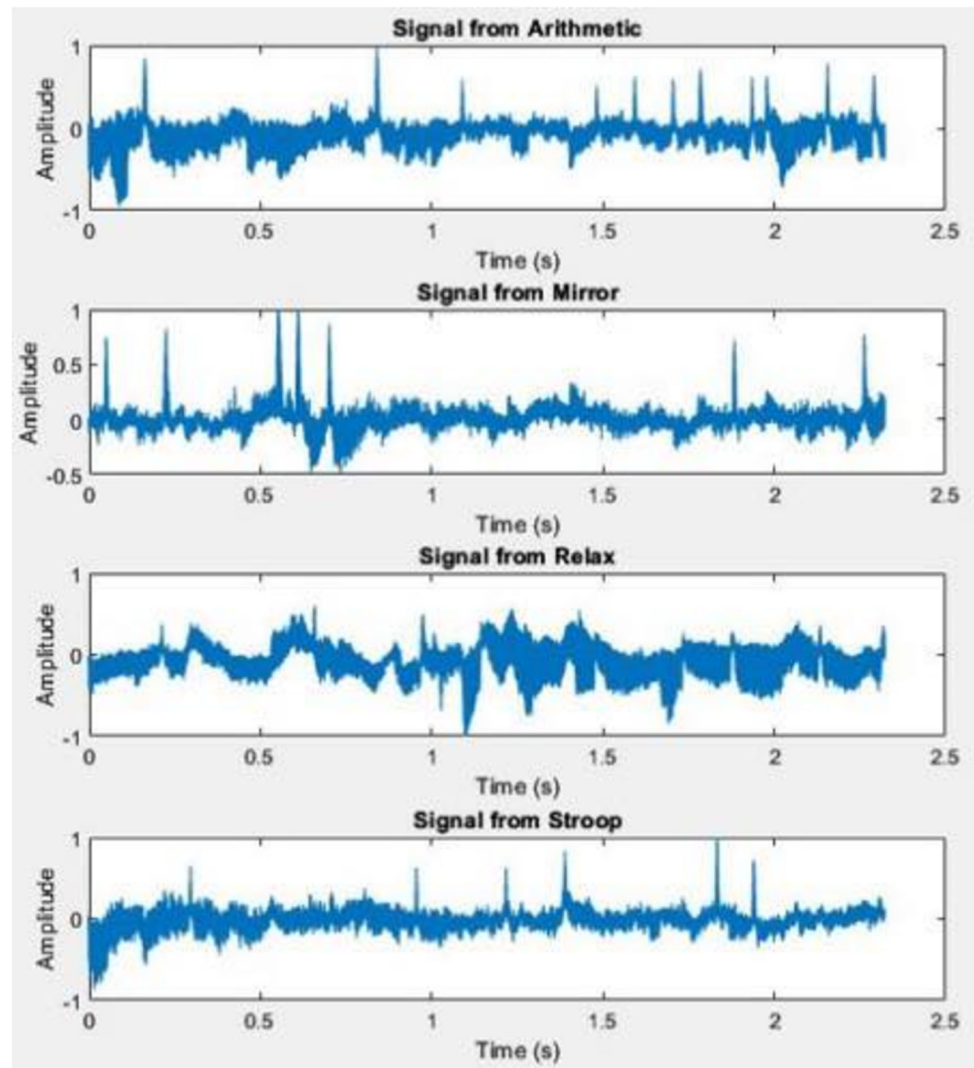
2.6 Mel spectrogram

Mel spectrogram [36] is a popular tool for analyzing audio signals, to record the distinct frequency and intensity patterns found in each signal. Mel spectrogram is based on an output of the audio signal *generated* from the DFT- and Mel-scaled filter bank.

Mel spectrogram is employed to convert audio segments from time–frequency domain signals to spectrogram images. The audio signal's time–frequency representation is the outcome of the Mel spectrogram. The Mel spectrogram's variables are the number of Mel parameters, the kind of window, and the length of the Mel spectrogram.

Mel spectrogram is generated in the domain of time–frequency which includes time and frequency information about the audio signal. It displays the spread of signal energy on the Mel-scale frequency, which is comparable to

Fig. 3 EEG contraction from SAM 40 database with 32 channels to one channel for four stress tasks



human auditory perception, by transforming the linear frequency range into a logarithmic one. Mel spectrogram displayed the changes in audio signal over time and employed the decibel scale for color coding. To extract a Mel spectrogram, we convert the frequency into a band with a Mel-scale by short-time Fourier transform (STFT) [48] as shown in Eq. 5

$$MS(x) = \text{Spectrogram}(x) \odot Bm(k) \quad (5)$$

where Bm represents filter banks, m denotes the overall number of filters, and spectrogram (x) is the results taken from STFT.

Then, the spectrogram images were segmented into small windows with $96 \times 64 \times 1$ L, where 96 denotes the parameters of Mel spectrograms, 64 denotes the Mel bands, and L denotes the length of the Mel spectrogram. In this study, the filter bank coefficients are applied to the Mel spectrogram to plot spectrogram images for four stress tasks as shown in Fig. 4.

2.7 Classification

The output of 64,800 Mel spectrogram segmented images derived from 480 signals in the SAM 40 database is applied to the training network. In the classification step, all 64,800 images were split up into three categories: 70% for training, 15% for validation, and 15% for testing as shown in Table 3. The proposed model was built using the VGGish-CNN structure for stress classification. The classification process consists of the VGGish network [49] as feature extraction and CNN [50] as a classifier. The extracted features are 4096 for each segmented image. The proposed model is based on relevant features for stress detection without manually selected features. CNN can be used to classify stress using the retrieved features as input.

In the last research [49], the VGGish network is a deep learning model trained on YouTube audio data, resulting in relatively high performance in speech detection algorithms. The VGGish network is made up of 24 layers including

three fully connected layers, convolutional, ReLU, and max-pooling layers [33].

In the proposed model, we removed the three last layers from the pre-trained VGGish network and added instead CNN layers to perform the classification. The VGGish with 21 layers is firstly built to identify features from the EEG signals for stress detection as in Fig. 5. The CNN consists of 27 layers including convolution layers, ReLU layers, normalization layers, max-pooling layers, and fully connected layers as shown in Fig. 6. The fully connected layers with 1000, 500, and 20 neurons are used. Then, a fully connected layer is coupled to a softmax layer which is followed by a classification layer for producing the classification results. Finally, the classification performance is evaluated using two models including the VGGish-CNN structure and VGGish classifier based on test sets.

Table 3 Dataset division for the classification process

Dataset	No of images $\times$ No of features
Training set	45,360 $\times$ 4096
Validation set	9720 $\times$ 4096
Test set	9720 $\times$ 4096
Total	64,800 $\times$ 4096

3 Results

3.1 Experimental setup

The proposed model is implemented by MATLAB R2022b software on an operating platform with a core i7 processor and sixteen gigabytes of RAM. After the conversion of the dataset to spectrogram segmented images, the ratio of the

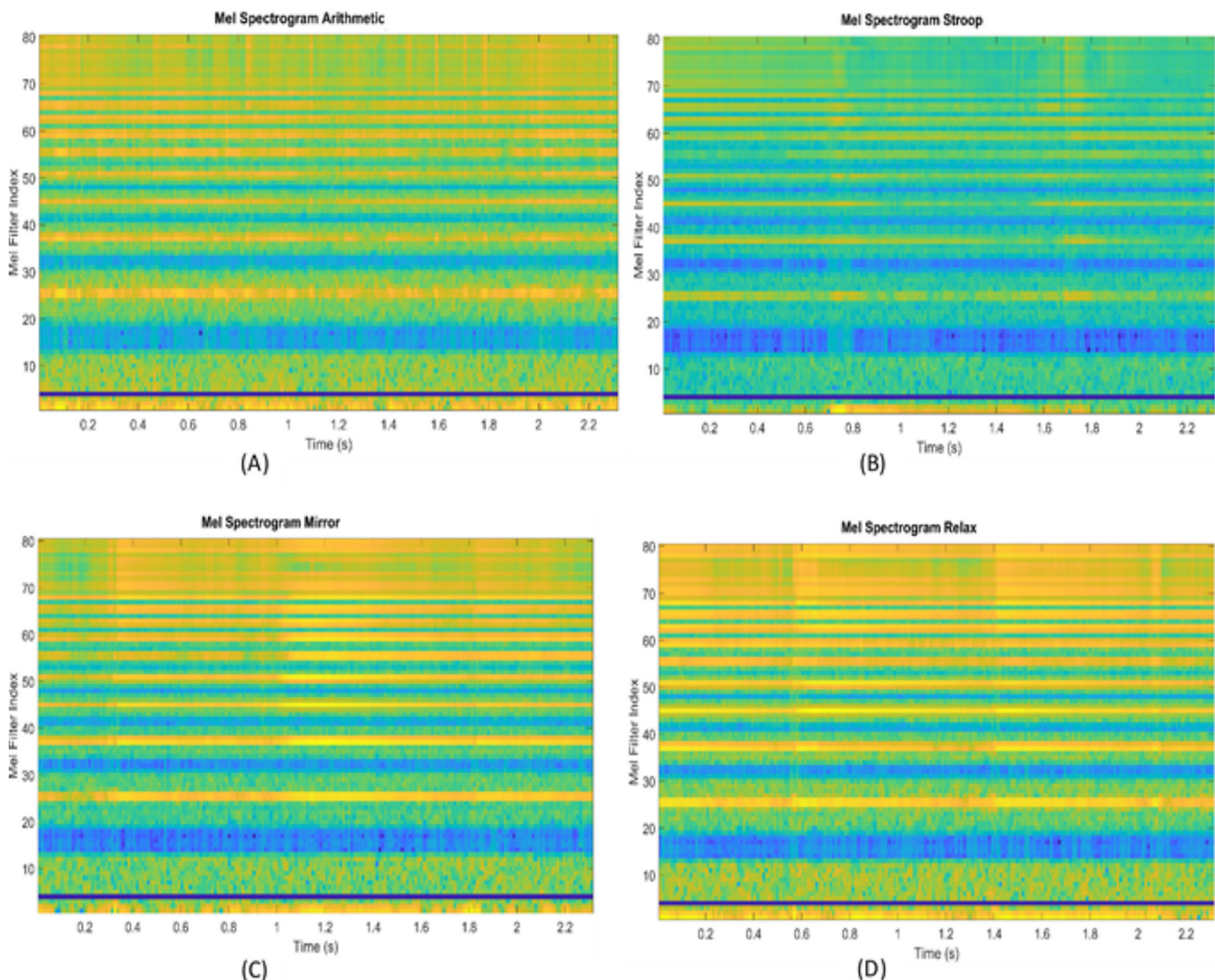


Fig. 4 Mel spectrogram of segmented signals for four stress tasks: **A** arithmetic, **B** SCWT, **C** mirror image recognition, and **D** relaxing

training set is 70%, the validation set is 15%, and the test set is 15%. Considering the network training, the learning rate was set to 0.0001, the batch size was 15, and 100 epochs were used by the Adam optimizer. In this study, we selected the cross-entropy (CE) function as the objective function for network training to calculate the loss between the actual features and anticipated features as shown in Eq. 6.

$$CE = \sum_{k=1}^k -p_k \log(p_k), P_k \in (0, 1) \quad (6)$$

where p represents the true distribution and K represents a one-hot vector of length.

In the second experiment, the k-fold cross validation process [51] is used to prove the resilience of the VGGish-CNN model. In this paper, we used fivefold cross validation that separated the database into five subsets of equal cases with 80% for training set and 20% for test set.

3.2 Performance metrics

Generally, the performance of stress detection from EEG signals is evaluated using five public metrics [52]. Performance metrics are standard tools for any classification model to facilitate comparison between different models.

In the proposed model, the VGGish is used to extract 4096 features from Mel spectrogram images and use CNN as a classifier to produce multi-classification for four stress tasks from the SAM 40 database.

The model training is implemented using 45,360 and 9720 images for training sets and validation sets, respectively, as shown in Table 4.

For the model evaluation, five values that are accuracy, precision, sensitivity, specificity, and F1-score employed on the output of the CNN classifier using 9720 test images.

$$\text{Accuracy} = \frac{\sum_{i=1}^c \frac{TP_i + TN_i}{TP_i + FN_i + FP_i + TN_i}}{c} \quad (7)$$

$$\text{Sensitivity} = \frac{\sum_{i=1}^c \frac{TP_i}{TP_i + FN_i}}{c} \quad (8)$$

$$\text{Specificity} = \frac{\sum_{i=1}^c \frac{TN_i}{TN_i + FP_i}}{c} \quad (9)$$

$$\text{Precision} = \frac{\sum_{i=1}^c \frac{TP_i}{TP_i + FP_i}}{c} \quad (10)$$

$$F1 - \text{score} = 2 \frac{\text{Precision} * \text{Sensitivity}}{\text{Precision} + \text{Sensitivity}} \quad (11)$$

where c is four stress tasks. TP refers to true positive for correctly diagnosed stress, TN refers to true negative for correctly diagnosed stress, FN refers to false negative for incorrectly diagnosed stress, and FP refers to false positive for erroneously diagnosed stress.

Also, the confusion matrix and receiver operating characteristics (ROC) curve are used to assess the suggested model's performance. ROC curve visualized the relation between the sensitivity and 1-specificity to provide the benefit of being independent of the class distribution.

3.3 Comparative analysis

The EEG recordings of the students from the SAM 40 database [32] are preprocessed and converted into a time–frequency domain based on Mel spectrograms, which feed VGGish [36], a feature extractor followed by a CNN classifier. For the preprocessing step, EEG contraction from the SAM 40 database with 32 channels to one channel for four stress tasks is applied to 480 signals. For the creation of Mel spectrogram images, a sliding window with a duration of 25 ms was applied to divide the original 3200 data points into small segments, with an overlap of 99% of the window's length and a window size of 512 to obtain 2D images as in Fig. 4.

This study investigated feature extraction using the VGGish pre-trained model in the framework of stress categorization from the SAM 40 database. The goal is to

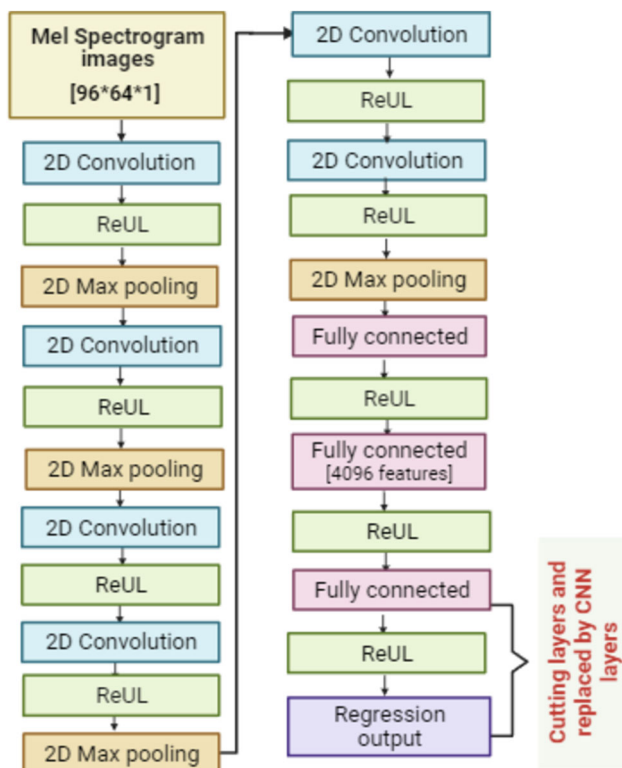


Fig. 5 VGGish architecture as feature extraction for stress detection model on SAM 40 database

Fig. 6 CNN architecture as a classifier for stress detection model on SAM 40 database

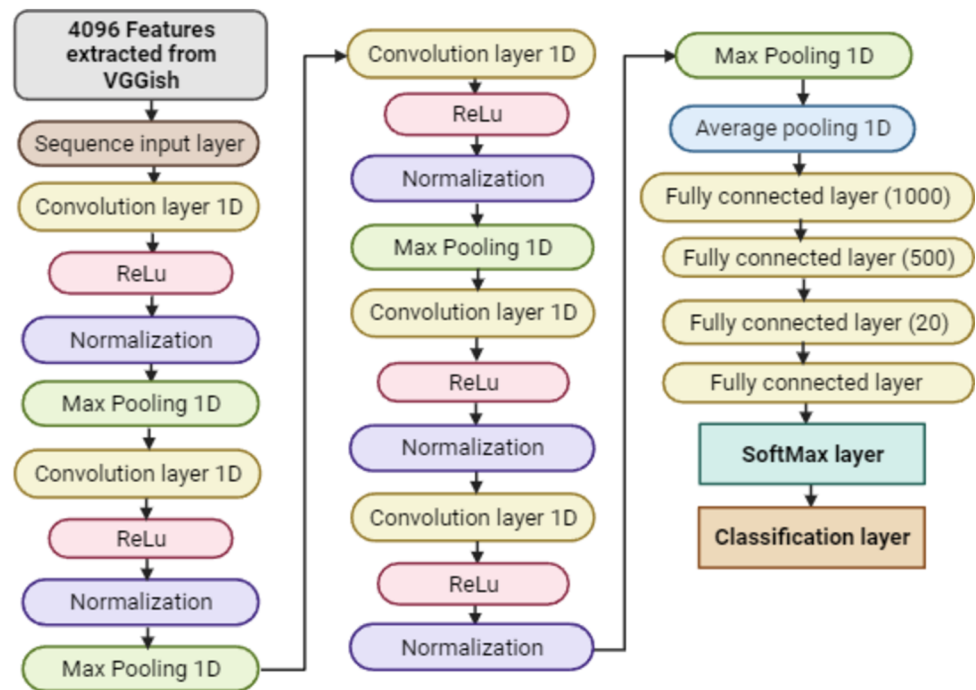


Table 4 Experimental results of two proposed models using five performance metrics on the SAM 40 database

Proposed model	Accuracy %	Sensitivity %	Specificity %	Precision %	F1-score %
VGGish-CNN	99.25	99.25	99.75	99.25	99.25
VGGish	22.84	22.84	74.28	20.48	20.60

determine how well VGGish extracts pertinent features that can be used to improve EEG signal quality and decrease noise interference. We compare the efficiency of the VGGish-CNN model and the VGGish model only to highlight the high classification accuracy for stress detection. The training progress of the VGGish-CNN model on the SAM 40 database is applied in Fig. 7.

For the model with CNN, it used the VGGish-CNN model including 21 layers from VGGish and 27 layers from CNN as shown in Figs. 5 and 6. This model showed high performance, with an accuracy of 99.25%, sensitivity of 99.25%, specificity of 99.75%, precision of 99.25%, and F1-score of 99.25% as shown in Table 4.

For the model without CNN, it still used the VGGish only with 24 layers as a classifier. This VGGish classifier showed low efficiency, with an accuracy of 22.84%, sensitivity of 22.84%, specificity of 74.28%, precision of 20.48%, and F1-score of 20.60%.

Figure 8 displays the confusion matrix of the VGGish-CNN model for stress recognition computed on test sets from the SAM 40 database. The result of the confusion matrix by the VGGish-CNN model improved the accuracy for all stress tasks. The diagonal parts in the confusion matrix represent the correctly classified four stress tasks. The accuracy rates of arithmetic, SCWT, mirror image, and

relaxing are 99.7%, 99.6%, 99.6%, and 98.1%, respectively. The stress recognition model based on VGGish-CNN is proposed, which combines the VGGish network with CNN to effectively improve the recognition accuracy of the EEG signals on the SAM 40 database.

As shown in Fig. 9, the ROC curve of the two proposed models reported the superiority of the VGGish-CNN model rather than the VGGish model. The EEG signal was more thoroughly represented by the VGGish features, which improved the ability to distinguish between the four stress tasks from the SAM 40 database.

For comparative analysis, the StressNet [53] is a recent stress detection architecture from EEG signals based on CNN and LSTM networks that yielded an accuracy of 97.8% on the combination of DEEP and SEED datasets. The previous studies on stress detection proposed SVM [31] and CNN [54] with an accuracy of 92.3% and 96%, respectively. It means that the proposed model outperforms these other studies, as shown in Table 4.

3.4 Validation process of VGGish-CNN model

The k-fold cross validation [51] is a more suitable process for validating a stress classification model using the VGGish-CNN on the SAM 40 database and applying the

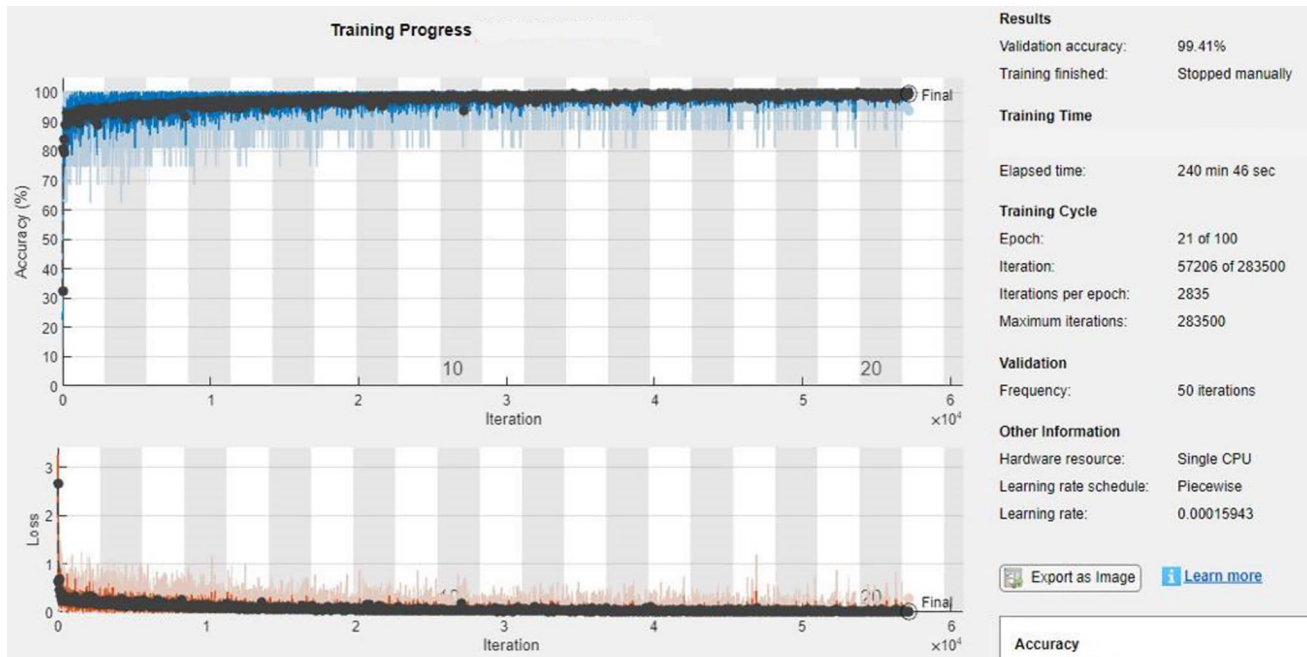


Fig. 7 Training progress for VGGish-CNN model on SAM 40 database

		Accuracy 99.25%			
		Arithmetic	Mirror	Relax	Stroop
Output Class	Arithmetic	99.7% 24.8	0.2% 5	0.7% 16	0.2% 5
	Mirror	0.1% 3	99.6% 23.8	1.2% 30	0.2% 4
	Relax	0.0% 1	0.0% 0	98.1% 23.0	0.0% 0
	Stroop	0.1% 3	0.2% 5	0.0% 1	99.6% 24.5
		Target Class			

Fig. 8 Confusion matrix of VGGish-CNN model on SAM 40 database

performance metrics on each fold. This process was executed 5 times, and each subset was divided into 80% of the training set and 20% of the test set. Table 5 shows the performance of the VGGish-CNN model using one–five-fold cross validation on the SAM 40 database.

The validation results confirmed the accuracies are 97.97%, 97.29%, 97.12%, 99.35%, and 98.74% for one–fivefold cross validation, respectively. The high validation metrics of the VGGish-CNN model were achieved when evaluated by fourfold cross validation. This fourfold cross validation showed high performance, with an accuracy of 99.35%, sensitivity of 99.36%, specificity of 99.79%,

precision of 99.36%, and F1-score of 99.36% as shown in Table 5. Therefore, the validation findings indicated that the VGGish-CNN model can indeed generalize well to unseen subjects.

4 Discussion

In modern life, stress is becoming more prevalent in many situations, and its diagnosis is essential [55]. EEG signal is an important tool for BCI applications because it has extensive knowledge about mental states and the brain's

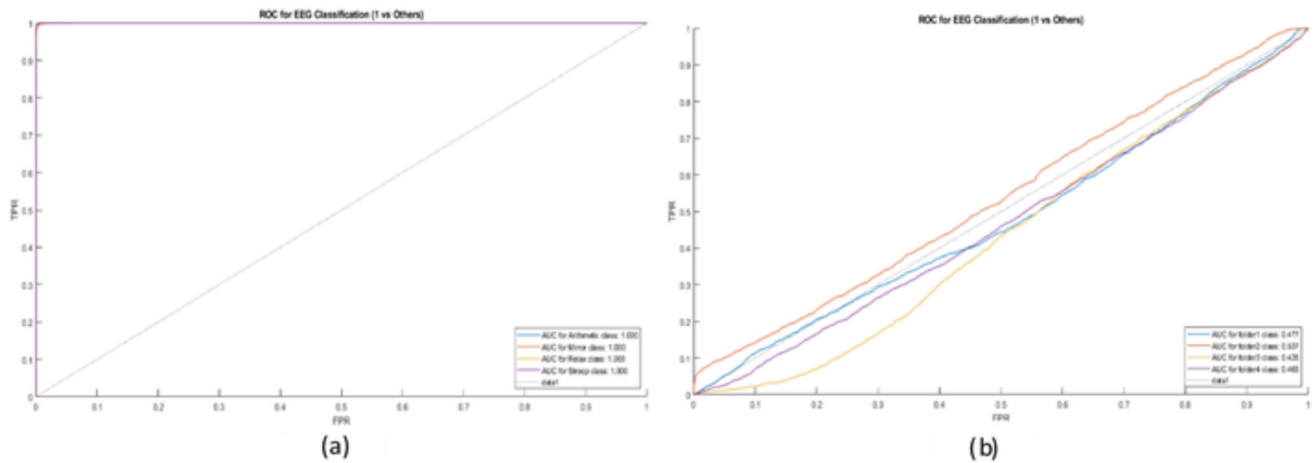


Fig. 9 ROC curve for two proposed models: **a** VGGish-CNN model, **b** VGGish model

electrical activity [3]. Several studies [56] have analyzed EEG signals under stress tasks by applying machine learning and deep learning algorithms on different databases.

Recently, the SAM 40 database [32] has recorded EEG signals with 32 channels along different short-term stress tasks, which are induced SCWT, answering mathematical problems, finding mirror-identical images, and relaxing.

In this study, signal preprocessing, segmentation, filtering, and spectrogram play an influential role in the success of classifier techniques because they help in creating a supporting tool for EEG-based stress detection as shown in Fig. 1. After these pre-steps, the proposed VGGish-CNN model is executed on the SAM 40 database, combining the 4096 features from the VGGish [56] and classifying them using the CNN structure.

The preprocessing step of the EEG signal from the SAM 40 database involves normalization and construction in one channel for lowering noise, eliminating unnecessary channels, and lowering error signals. EEG preprocessing is essential as an initial step before the EEG classification because original noisy data could make training challenging and delay the network learning. For the EEG categorization assignment, the unprocessed data took several

seconds during training and affected the results [41]. The EEG segmentation step based on DFT [34] is effective for improving findings and for identifying which portions of the recorded data have the highest levels of brain activity associated with stress. After that, the Mel-scaled filter bank is applied to allow a better resolution at low frequencies and less at high frequencies. It is used in gathering energy at each important frequency band and nearly matches the spectrum form. Therefore, it aids in facilitating the harmonic framework. Audio signals passed through a Mel-scaled filter bank to produce Mel spectrogram images. Then, 64,800 Mel spectrogram images are fed into a stress recognition system to apply the classification task for the EEG signals. To match the VGGish [47] input data, the obtained spectrograms are divided into K-segmented images, each image has $96 \times 64 \times 1$ pixels and K represents the amplitude value divided by 96. The Mel segmented spectrogram images were separated into a 70% training set, a 15% validation set, and a 15% test set to perform the stress classification from EEG signals. For validation process, the spectrogram images were divided into short overlapping segments and then randomly divided into 80% training and 20% test to apply the k-fold cross validation process [51].

Table 5 Performance of VGGish-CNN model using k-fold cross validation on SAM 40 database

Item	One fold	Two fold	Three fold	Four fold	Five fold
Accuracy %	97.97	97.29	97.12	99.35	98.74
Sensitivity%	97.98	97.29	97.13	99.36	98.74
Specificity%	99.33	99.10	99.04	99.79	99.58
Precision%	98.00	97.34	97.18	99.36	98.74
F1-score%	97.98	97.28	97.13	99.36	98.74

Bold values indicate the best results

For training progress, the validation accuracy achieved 99.41% at 21 epochs as shown in Fig. The results revealed that the VGGish-CNN model significantly improved the stress classification efficiency compared to the VGGish model only. The choice of feature extraction for the classification affected the accuracy and training time, but in some cases feature extraction provides better results than training a network without features as in the suggested model. The main benefit of the suggested CNN model is to automate feature extraction, instead of manual features as in traditional machine learning models [27, 28]. This SAM 40 database is not suitable for traditional models that use manual feature extraction techniques before the classification process due to its size.

Therefore, the VGGish-CNN model obtained better results than VGGish with the SAM 40 database. The k-fold cross validation is used to validate the stress classification using the VGGish-CNN model on the SAM 40 database. The k-fold cross validation is easy to use and typically yields a less biased and less overfitting model than other methods.

However, the execution time of the VGGish-CNN model is 184 s for short-term stress detection on the SAM 40 database. In this proposed model, the reason for the high execution time is related to 4096 features and the CNN classifier.

To this end, the combination of the CNN classifier and the VGGish feature extraction provides novel solutions to tackle problems in the short-term stress detection field based on EEG signals.

The drawbacks of research on EEG-based stress detection depend on the inability to comparison in different techniques utilized to gather and interpret EEG databases such as different channels, sampling rates, filtering strategies, and preprocessing procedures [57]. Another drawback is the unavailability of large samples in the EEG database for stress analysis. In many studies, there are small stress samples from EEG databases because the gathering and analyzing of this type of database can be time-consuming and expensive. Therefore, a limited number of clinical cases to create a database impacted on results' generalizability and caused the model to become overfit to the particular dataset being utilized.

The limitations of the proposed model are that the SAM 40 database has been overly limited by small samples of only 40 students and a small number of channels. Also, there is no comparative analysis with previous studies because this study is the preliminary use of stress classification based on EEG signals on the SAM 40 database. The proposed model takes a long execution time for stress classification due to the interactions between the VGGish and CNN architectures.

Despite the positive results of the tested VGGish-CNN model, we recommend further research on feature extraction for stress detection.

In the future, we strongly advise researchers to gather stress signals from EEG recordings in a quiet, controlled environment with enough time to fit the electrodes to each participant.

5 Conclusion

The suggested model demonstrates the efficacy and promise of the VGGish-CNN model for stress detection based on EEG signals from the SAM 40 database. The extracted features from the VGGish model offer a useful foundation for enhancing EEG signal quality, lowering disturbance from noise, and identifying the differences between four stress tasks. This discovery opens up new possibilities for future advancement in the field of neuropsychological disease enhancement and advanced stress classification. Therefore, this proposed model will be used to help psychologists in neuropsychological clinics to identify stress levels in actual time. It can be extended to recognize the long-term stress of multitasks while retaining high accuracy and early prediction.

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Data availability The SAM 40 database is available freely and is hosted publicly at this link. <https://doi.org/https://doi.org/10.6084/m9.figshare.14562090.v1>.

Declarations

Conflict of interest The authors declare that there is no conflict of interest.

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